

FOUR-ITERATE COLLAPSE AND EXACT FIBRES FOR TRANSPOSE-ROW-COMPRESSSION DYNAMICS

ANONYMOUS

ABSTRACT. Left-compress every row of a labelled binary square matrix and transpose the result, then feed the output back into the same rule. We determine the complete functional graph. If r is the labelled row-sum vector, successive column-height vectors are r , its threshold conjugate r^* , and its decreasing rearrangement $r^\downarrow$; hence $F^4 = F^2$. The recurrent states are exactly the Ferrers matrices, where F acts by partition conjugation. There are $\binom{2n}{n}$ recurrent states, 2^n fixed states, and $(\binom{2n}{n} - 2^n)/2$ strict two-cycles. We give the exact three depth populations, including a weighted-partition formula for depth at most one. Independently, we determine every time-one and time-two target fibre, with zero-fibre criteria, and obtain image sizes $(n+1)^n$ and $\binom{2n}{n}$. The one-dimensional boundary is entirely fixed. Ferrers diagrams, conjugation, and line-sum theory receive no contribution credit; the retained iterative conjunction is OWNER_AMBER / HOLD_EXTERNAL.

1. THE LITERAL MAP AND THE SUBTRACTION BOUNDARY

Fix $n \geq 1$, write $[n] = \{1, \dots, n\}$, and let $\mathcal{X}_n = \{0, 1\}^{n \times n}$. Rows and columns retain their displayed labels. For $A \in \mathcal{X}_n$, put

$$r_j(A) = \sum_{k=1}^n A_{jk}, \quad r(A) = (r_1(A), \dots, r_n(A)).$$

The transpose-row-compression map is

$$(1) \quad F(A)_{ij} = \mathbf{1}\{i \leq r_j(A)\}, \quad i, j \in [n].$$

Thus one first moves every 1 in source row j to the left without changing that row's sum, and then transposes. Equivalently, source row j becomes an initial column of height $r_j(A)$. The rule is synchronous, deterministic, and total on the labelled carrier.

Binary matrices with prescribed line sums and generalized conjugate sequences are classical [3, 4]; Ferrers matrices and bigraphs are established objects [2]; and partition conjugation and diagonal hooks are standard [1]. We assign all of that material, as well as generic finite-map bookkeeping, zero contribution credit. The scoped object here is only the repeated literal map (1) and the conjunction of its exact clock and target-local inverse laws. A bounded owner search did not locate that conjunction. This non-hit is not novelty or priority evidence, and external circulation remains on hold.

For a height vector $h = (h_1, \dots, h_n) \in \{0, \dots, n\}^n$, define

$$(2) \quad D(h)_{ij} = \mathbf{1}\{i \leq h_j\}, \quad h_i^* = \#\{j : h_j \geq i\} \quad (i \in [n]).$$

Write $h^\downarrow$ for the decreasing rearrangement of h . The transform h^* is always a partition in the $n \times n$ square, padded with zeros. Let $\mathcal{P}_n$ be the set of all such partitions.

2. THE FOUR-ITERATE NORMAL FORM

The distinction between the labelled vector h and its rearrangement is essential at the first epoch.

Lemma 1 (height calculus). *The map D is injective and*

$$(3) \quad r(D(h)) = h^*, \quad (h^*)^* = h^\downarrow.$$

For $\lambda \in \mathcal{P}_n$, the operation $\lambda \mapsto \lambda^$ is ordinary Ferrers conjugation and is an involution.*

Proof. The initial run of ones in column j recovers h_j , proving injectivity. Row i of $D(h)$ has a one precisely in the columns with $h_j \geq i$, which proves the first identity. Threshold counts do not depend on coordinate order, so $h^* = (h^\downarrow)^*$. Reflecting the Ferrers diagram of the partition $h^\downarrow$ across its diagonal proves the second identity and the final assertion. $\square$

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Theorem 2 (all-time normal form). *For $A \in \mathcal{X}_n$, let $r = r(A)$. Then*

$$(4) \quad F(A) = D(r), \quad F^2(A) = D(r^*),$$

$$(5) \quad F^3(A) = D(r^\downarrow), \quad F^4(A) = D(r^*) = F^2(A).$$

Consequently, for every $q \geq 1$,

$$F^{2q}(A) = D(r^*), \quad F^{2q+1}(A) = D(r^\downarrow).$$

Proof. Equation (1) is $F(A) = D(r(A))$. Apply Lemma 1 successively:

$$F^2(A) = D(r^*), \quad F^3(A) = D((r^*)^*) = D(r^\downarrow).$$

Since threshold counts ignore rearrangement, $(r^\downarrow)^* = r^*$, giving $F^4(A) = D(r^*)$. Reapplying F alternates the last two matrices. $\square$

For example, every matrix with row-sum vector $(1, 3, 0)$ follows the height trajectory

$$A \mapsto D(1, 3, 0) \mapsto D(2, 1, 1) \longleftrightarrow D(3, 1, 0).$$

Thus neither $F^2 = F$ nor $F^3 = F$ holds in general, despite the shallow universal bound.

3. RECURRENT STATES AND EXACT DEPTH LAYERS

Depth means distance to the recurrent set; a strict two-cycle is an orbit of two distinct states. Put

$$(6) \quad W_n = \sum_{\lambda \in \mathcal{P}_n} \prod_{i=1}^n \binom{n}{\lambda_i} = [z^n] \prod_{k=0}^n \frac{1}{1 - \binom{n}{k} z}.$$

The coefficient identity records the multiplicities of the $n+1$ possible parts.

Theorem 3 (complete functional graph). *The recurrent states are exactly $D(\lambda)$ for $\lambda \in \mathcal{P}_n$, and*

$$(7) \quad F(D(\lambda)) = D(\lambda^*).$$

Hence the recurrent, fixed, and strict-two-cycle counts are

$$(8) \quad \binom{2n}{n}, \quad 2^n, \quad \frac{1}{2} \left(\binom{2n}{n} - 2^n \right).$$

The exact depth sets are

$$\begin{aligned} L_0 &= \{D(\lambda) : \lambda \in \mathcal{P}_n\}, \\ L_1 &= \{A : r_1(A) \geq \dots \geq r_n(A)\} \setminus L_0, \\ L_2 &= \{A : r_i(A) < r_{i+1}(A) \text{ for some } i < n\}, \end{aligned}$$

with populations

$$(9) \quad |L_0| = \binom{2n}{n}, \quad |L_1| = W_n - \binom{2n}{n}, \quad |L_2| = 2^{n^2} - W_n.$$

For $n \geq 2$ the height is exactly two. For $n = 1$, both states are fixed, so the height is zero and the populations are $(2, 0, 0)$.

Proof. Theorem 2 implies that every $F^2(A)$ is fixed by F^2 . Conversely, if A lies on a cycle, F is invertible on that cycle, so cancelling F^2 from $F^4(A) = F^2(A)$ gives $F^2(A) = A$. The recurrent set is therefore $\text{Fix}(F^2)$. Every second image is $D(\lambda)$ for the partition $\lambda = r(A)^*$. Conversely, Lemma 1 gives $F^2(D(\lambda)) = D(\lambda)$, proving the characterization and (7).

Partitions in the square are in bijection with monotone boundary paths having n horizontal and n vertical steps, hence there are $\binom{2n}{n}$. A recurrent matrix is fixed precisely when $\lambda = \lambda^*$. The diagonal hook lengths of a self-conjugate shape form an arbitrary subset of $\{1, 3, \dots, 2n-1\}$, and this construction is reversible inside the square. Thus there are 2^n fixed states. Conjugation pairs all remaining shapes, which proves (8).

It remains to distinguish depths one and two. The first image $D(r(A))$ is recurrent exactly when the labelled row-sum vector is decreasing. For a fixed decreasing vector λ , the supports of its rows can be chosen in $\prod_i \binom{n}{\lambda_i}$ ways. Summing proves that W_n is the depth-at-most-one population and gives (9). A matrix with only $A_{1n} = 1$ has depth one when $n \geq 2$, whereas a matrix with only $A_{21} = 1$ has depth two. This proves sharpness and all boundary claims. $\square$

4. EVERY-TARGET FIBRES AT TIMES ONE AND TWO

The inverse laws retain labelled row assignment, which disappears from the forward partition quotient. For $\lambda \in \mathcal{P}_n$, let $m_k(\lambda)$ be the multiplicity of part k , including $k = 0$.

Theorem 4 (complete inverse atlas). *For an arbitrary target $B \in \mathcal{X}_n$,*

$$(10) \quad |F^{-1}(B)| = \begin{cases} \prod_{j=1}^n \binom{n}{h_j}, & B = D(h), \\ 0, & B \notin D(\{0, \dots, n\}^n). \end{cases}$$

Consequently $|\text{im } F| = (n+1)^n$.

At time two, let $\mu \in \mathcal{P}_n$, put $\lambda = \mu^*$, and use zero for every non-Ferrers target. Then

$$(11) \quad |(F^2)^{-1}(D(\mu))| = \frac{n!}{\prod_{k=0}^n m_k(\lambda)!} \prod_{k=0}^n \binom{n}{k}^{m_k(\lambda)}.$$

Thus $|\text{im } F^2| = \binom{2n}{n}$. Both formulas include the all-zero and all-one height boundaries and remain valid for $n = 1$.

Proof. The equation $F(A) = B$ is possible precisely when every column of B is an initial segment. In that case injectivity of D gives $B = D(h)$ uniquely and forces source row j to have sum h_j . Its support has $\binom{n}{h_j}$ choices, independently across labelled rows. This proves (10); arbitrary h gives the first image count.

By (4), $F^2(A) = D(\mu)$ is equivalent to $r(A)^* = \mu$. Lemma 1 says equivalently that the decreasing rearrangement of $r(A)$ is $\lambda = \mu^*$. Assigning this multiset of parts to the labelled rows gives $n! / \prod_k m_k(\lambda)!$ distinct row-sum vectors. For each assignment, independent row supports contribute $\prod_k \binom{n}{k}^{m_k(\lambda)}$. This proves (11). Finally, every partition μ occurs, and no other second target occurs, so the boundary-path count gives $|\text{im } F^2| = \binom{2n}{n}$. $\square$

Summing either fibre law over its admissible targets recovers 2^{n^2} . This mass identity is a check, not a derivation of the target-local formulas.

5. EXACT CONTROL, LIMITATIONS, AND DECLARATIONS

The paper-local standard-library verifier is independent of the scouting implementation. It exhausts all 2^{n^2} matrices for $1 \leq n \leq 4$, constructs literal transitions entry by entry, and compares every target fibre with (10)–(11). Separate partition-level controls reach $n = 12$. Its complete boxes are:

n	states	$ \text{im } F $	$ \text{im } F^2 $	fixed	strict 2-cycles	$ L_1 $	$ L_2 $
1	2	2	2	2	0	0	0
2	16	9	6	4	1	5	5
3	512	64	20	8	6	164	328
4	65,536	625	70	16	27	10,051	55,415

The audit also searches explicitly for and finds counterexamples to the false strengthenings $F^2 = F$ and $F^3 = F$, distinguishes labelled row order from its multiset, attacks columns with holes, and checks the $n = 1$ boundary. It makes 5,336,613 exact assertions. Finite computation is falsification pressure, not proof or ownership evidence.

The theorem depends on a square carrier, fixed labels, synchronous updating, and transposition after compression. Rectangular variants, sorting inside the literal update, asynchronous schedules, random kernels, and unlabelled quotients are outside scope. The bounded owner search leaves live ownership risk; status is OWNER_AMBER / HOLD_EXTERNAL, and no novelty, priority, or circulation claim is authorized. Data availability. No external data were used. The exact verifier and canonical transcript accompany the source.

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